
Using Cluster Managers

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Two cluster managers are supported by IC Validator scripts:

- [icvgrid](#)
- [icvlsf](#)

GRID Queuing Method for IC Validator

The `icvgrid` script allows you to submit distributed jobs to a Linux cluster using the GRID queuing method. This method uses the Sun Grid Engine (SGE).

Running Icvgrid Script

`icvgrid` is used in place of the `icv` executable. The `icv` command-line options are valid when using `icvgrid` in addition to the command-line options listed below.

To run `icv` using the GRID queuing method:

```
% icvgrid [options] runset
```

Icvgrid Command-Line Options

Option	Definition
-h	Prints a <code>icvgrid</code> usage message.
-dpN	Number of slave processes.
-mem64=P	Number of processes with 64 GB dedicated memory.

-mem32=Q	Number of processes with 32 GB dedicated memory.
-mem16=R	Number of processes with 16 GB dedicated memory.
-mem8=S	Number of processes with 8 GB dedicated memory.
-mem4=T	Number of processes with 4 GB dedicated memory.
-mem2=U	Number of processes with 2 GB dedicated memory.
-amd64	Use amd64 machines only.
-ia32	Use intel32 machines only.
-ia64	Use intel64 machines only.
-V	Print script version.
-cmd	Print full command line and arguments.
-wait	Wait for icv to finish before exiting.
-redhat	Work around Red Hat NUMA bug.
-noLocalDisk	Do not use local cache directories.
-cleanup	Remove all scratch directories for this user.
-l	User define resources; for example; -l arch=glinux,cputype=amd64.
-quick	Quick start icv. Do not wait for all hosts to queue up before running.

-P	User define project such as bnormal.
-notimer	Stop the 5 minute timer used for busy cluster. This will have unlimited wait time.
-igcae	sets up default configuration for igcae cluster: -P bnormal -l arch=glinux, mem_free=1G, mem_avail=1G, cputype=amd64 - For example: icvgrid -igcae -dp2 runset.ev
-conf <i>path</i>	Local path for the SGE configuration resources file. The default setting is: /remote/sge/cells/common/settings.sh. If the SGE resources are not found, Icvgrid issues a message and stops.

Selecting Memory Resources

icvgrid allows you to select the processors required for a distributed job, based on the memory requirements for that job. The machines on the cluster are configured with 64, 32, 16, 8, 4 and 2 GB of memory. icvgrid dedicates the required memory for each processor and stops other processes from accessing that memory.

The number of processors needed for a job is specified with the `dpN` option and memory requirements are specified with the `-mem` options. If the memory requirement is not specified for all processors, icvgrid defaults the unspecified processors to 2 GB of memory.

Use the UNIX command `qstat` to view the current jobs in progress.

Example 1: This job asks for four processors, each having exclusive use of 16 GB of memory. icvgrid forces the selection of machines that have at least 16 GB of memory for a single processor.

```
% icvgrid -dp4 -mem16=4 runset.ev
```

For example the machines selected are, with only one processor per machine is used:

Hplc06-p
Hplc07-p
Hplc08-p
Hplc09-p

Example 2: This job needs 32 GB in order to accomplish memory intensive preprocessing. Several 16GB processors are needed for running the runset. In addition, 32 processors with at least 4 GB each are needed to complete the 32 partition data portion of the run.

```
% icvgrid -dp32 -mem32=1 -mem16=5 -mem4=28 runset.ev
```

The machines available are listed below. Note that the user requested five 16GB machines but only four are available, and requested 28 4GB machines, but 31 are available in the grid. Once the 32GB machine is used for the memory intensive preprocessing, icv continues to use it.

Hplc40-p	32 GB
Hplc02-p	16 GB
Hplc04-p	16 GB
Hplc05-p	16 GB
Hplc06-p	16 GB
Hplc07-p	4 GB
Hplc11-p	4 GB
Hplc11-p	4 GB
Hplc12-p	4 GB
Hplc12-p	4 GB

...

Hplc25-p 4 GB

Hplc25-p 4 GB

Using a Local Disk

icvgrid automatically configures the temporary SCRATCH disk space for the slave machines. Typically, when using distributed processing you would specify `group_cache_dir` to where local disk space is available for the slave process. icvgrid does this for you by creating a temporary directory under SCRATCH using your ID and the icvgrid process ID, thereby creating unique, temporary disk space for each job.

While the job is running, you can view the contents of the `group_cache_dir` on each machine, such as `/SCRATCH/myname/group_25109`. When the job is finished, the directory is removed.

Submitting Jobs With icvgrid Script

You can submit jobs to the cluster with icvgrid. If icvgrid cannot obtain the processor resources at the time of submission, the script waits five minutes then resubmits the job and tries to obtain the resources a second time. This process continues until icvgrid obtains the resources or until you terminate the job.

Load Sharing Facility (LSF) for IC Validator

The `icvlsf` script uses Platform™ Computing Corporation LSF® (Load Share Facility) software for DRM (Distributed Resource Management). The integration between LSF and IC Validator automates the process of launching distributed IC Validator runs.

Requirements

Before running icv in distributed mode with LSF, in addition to a distributed processing license the following are required:

- LSF profile that contains all the necessary environment variables to run in LSF mode. An example path for the profile is `/global/cust_apps/lsf/conf.us54rtp/profile.lsf`. You can specify the profile using the `$LSF_ENVDIR` environment variable.
- Any site-specific settings needed for the `icvlsf` script to run properly. Use the command-line options to set the site-specific settings.

Setting	Definition
-cN	Selects the method used to start jobs on remote hosts: • 0 for <code>lsrun</code> • 1 for <code>rsh</code> • 2 for <code>ssh</code> • 3 for <code>bsub</code> (default) For example, <code>-c3</code> issues <code>bsub</code> on the command line: <pre>%icvlsf -c3 -dp4 runset.ev</pre>
-R	Specifies extra resource requirements, such as a particular machine type or platform type. For example, to set the platform to <code>emt64</code> or <code>amd64</code>: <pre>%icvlsf -R "emt64 amd64" -dp4 runset.ev</pre>
-q	Specifies a non-default queue, if needed. The <code>icvlsf</code> script runs in interactive mode and needs a queue to handle this mode. For example, to use the <code>iheavy</code> interactive queue: <pre>%icvlsf -q iheavy -dp4 runset.ev</pre>

Running icvlsf Script

`icvlsf` is used in place of the `icv` executable. The `icv` command-line options are valid when using `icvlsf` in addition to the LSF command-line options listed below.

To run `icv` using the LSF queuing method:

```
%icvlsf [options] runset
```

icvlsf Command-line Options

Option	Definition
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-h	Prints a usage message.
-dpN	Number of slave processes.
-mem64=P	Number of processes with 64 GB dedicated memory.
-mem32=Q	Number of processes with 32 GB dedicated memory.
-mem16=R	Number of processes with 16 GB dedicated memory.
-mem8=S	Number of processes with 8 GB dedicated memory.
-mem4=T	Number of processes with 4 GB dedicated memory.
-V	Prints the icvlsf script version.
-cN	selects the method N used to start jobs on remote hosts: 0 for lsrn, 1 for rsh, 2 for ssh, 3 for bsub
-q <name>	Specifies non-default queue such as "iheavy".
-R "resource"	Specifies extra resource requirements: ex. "maxmem>7112"
-nolog	Stops the log from scrolling on the terminal.
-dprc	Creates and uses dprc file to run job.
-allcpu	Creates and uses dprc file to run job.
-wait	Waits for icv to finish before exiting

-fast	All CPUs specified in the -dp and mem options are assigned, whether used or not. Without the fast option, a wait is instituted for each CPU to become busy prior to adding another CPU for work.
-noLocalDisk	Do not use local disk for group directory.
-quick	Do not wait for all hosts to queue before starting icv.
-x	Prompts for use of exclusive queues. Note this option must include an enabled LSF queue for exclusivity in the lsf.conf and lsb.params.

Selecting Memory Resources

icvlsf allows you to select the processors required for a distributed job based on the memory requirements for that job. The number of processors needed for a job is specified with the dpN option and memory requirements are specified with the -mem options. If the memory requirement is not specified for all processors, icvlsf defaults the unspecified processors to 4 GB of memory.

During the run, a wait is instituted for each CPU to become busy prior to adding another CPU for work. If the number of CPUs specified in the -dp option is not available, icvlsf continues checking for an available host until it is successful. However, if at least one CPU is available, the run continues. As more CPUs become available, the script continues adding the CPUs to the run until the maximum number specified is reached. See -fast icvlsf command-line option to assign all CPUs at the start of the run.

Example 1: The -dp option specifies a maximum of 4 CPUs to be used for the distributed icv run.

```
% icvlsf -dp4 runset.ev
```

Example 2: Submits a run with a maximum of 4 CPUs on hosts that have a minimum of 64 gigabytes of memory.

```
% icvlsf -mem64=4 -dp4 runset.ev
```

Example 3: Run with a maximum of 4 CPUs without waiting for each CPU to become busy before adding another CPU for work.

```
% icvlsf -dp4 -fast runset.ev
```

Using a Local Disk

icvlsf automatically configures the temporary SCRATCH disk space for the slave machines. Typically, when using distributed processing you would specify `group_cache_dir` to where local disk space is available for the slave process. icvlsf does this for you by creating a temporary directory under SCRATCH using your ID and the icvlsf process ID, thereby creating unique, temporary disk space for each job.

While the job is running, you can view the contents of the `group_cache_dir` on each machine, such as `/SCRATCH/myname/group_ 25109`. When the job is finished, the directory is removed.

This option can be turned off by using the `-noLocalDisk` option on the icvlsf command line.

```
% icvlsf -dp4 -noLocalDisk runset.ev
```

Using a Number of CPUs per Machine

If a number of CPUs are required per machine for a job run, the LSF resource `-R span[ptile=N]` option can be used on the command line:

```
% icvlsf -dp4 -R span[ptile=2] runset.ev
```

The above command will queue 2 machines with two CPUs to run the job.

Using Exclusive Queues

icvlsf script has the capability to run jobs on an exclusive set of machines in a queue by using the `-x` option. However, the LSF exclusive queue option must be configured in the LSF cluster.

```
% icvlsf -dp4 -x runset.ev
```

The modification of the preemption of backfill and exclusive jobs are mainly configured by the LSF admin. The following are the configuration notes to set up the exclusive queue.

File Parameter Syntax and description:

lsb.params PREEMPT_JOBTYPE PREEMPT_JOBTYPE=BACKFILL

- ◆ Enables preemption of backfill jobs.
- ◆ Requires the line PREEMPTION=PREEMPTABLE in the queue definition.
- ◆ Only jobs from queues with a higher priority than queues that define resource or slot reservations can preempt jobs from backfill queues.

PREEMPT_JOBTYPE=EXCLUSIVE

- ◆ Enables preemption of and preemption by exclusive jobs.
- ◆ Requires the line PREEMPTION=PREEMPTABLE or PREEMPTION=PREEMPTIVE in the queue definition.
- ◆ Requires the definition of LSB_DISABLE_LIMLOCK_EXCL in lsf.conf.

PREEMPT_JOBTYPE=EXCLUSIVE BACKFILL

- ◆ Enables preemption of exclusive jobs, backfill jobs, or both.

lsf.conf LSB_DISABLE_LIMLOCK_EXCL LSB_DISABLE_LIMLOCK_EXCL=y

- ◆ Enables preemption of exclusive jobs.
- ◆ For a host running an exclusive job:

- ❖ lshosts displays the host status ok.
- ❖ bhosts displays the host status closed.
- ❖ Users can run tasks on the host using lsruncmd or lsgruncmd.

To prevent users from running tasks during execution of an exclusive job, the parameter LSF_DISABLE_LSRUN=y must be defined in lsf.conf.

- ◆ Changing this parameter requires a restart of all sbatchds in the cluster (badadmin hrestart).

Add to the queue definition in lsf_queues:

- ◆ EXCLUSIVE = Y

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